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300116(37)

B. E. (First & Second Semester) Examination 2020

(Old Scheme)

(AEI, Bio Tech., Chem., Civil, CSE, Elect., EEE, EI, ET&T, IT, Mech., Mining, Metallurgy, Mechatronics Branch)

ENGINEERING MECHANICS

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Part (a) of each question is compulsory. Attempt any **two** from (b), (c) and (d) from question Q. 1, Q. 3 and Q. 5 and attempt any **one** from (b) and (c) from Que. 2 and Q. 4.*

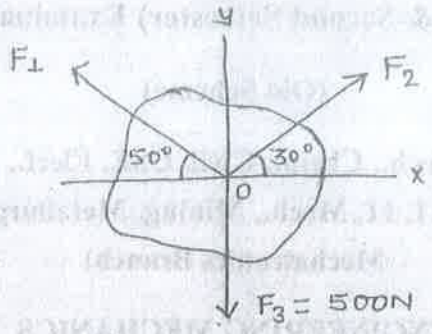
Unit-I

1. (a) Define force. What is the unit of force in S. I. system. 2

(b) Explain free body diagram in detail. 7

(c) A body acted upon by three forces F_1, F_2 and F_3

is in equilibrium. If the magnitude of the force F_3 is 500 N, find the forces F_1 and F_2. 7

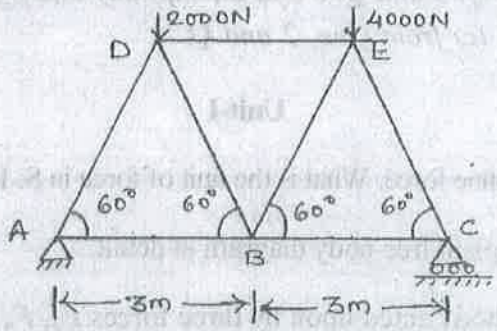


(d) State Law of triangle with proper diagram. 7

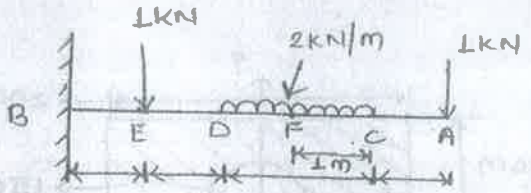
Unit-II

2. (a) Define truss. 2

(b) Using the method of joints, find the axial forces in all the memebers of a truss with the loading shown in figure below. 14



(c) Draw the shear force and bending moment diagrams for cantilever loaded as shown in figure below. 14



Unit-III

3. (a) Define limiting friction. 2

(b) State the law of dry friction. 7

(c) A wooden block rests on a horizontal plane. Determine the force required to pull it. Assume the mass m of the block to be 5 kg and the coefficient of friction $\mu = 0 \cdot 4$. 7

(d) Explain static friction and dynamic fricition. 7

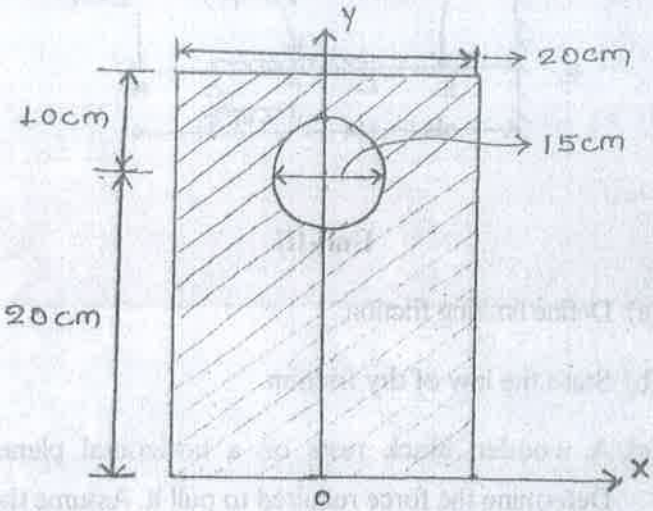
Unit-IV

4. (a) Define moment of inertia. 2

(b) Explain parallel axis theorem. 14

(c) Find the moment of inertia of a plate with a circular

hole about its centrioidal x-axis as shown in fig. below. 14



Unit-V

5. (a) State D'Alembert's principle. 2

(b) Explain principle of work and energy. 7

(c) Explain principle of Impulse and Momentum. 7

(d) A ball of mass M is dropped on to a spring of stiffness k from a height h. Find the maximum deflection δ of the spring. Assume $M = 5$ kg., $K = 500$ N/m $h = 10$ cm 7